

Week 4: Stochastic Optimal Control in Continuous Time

The general theory behind the worked example

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Learning Objectives

- ▶ Apply the dynamic programming principle in continuous time
- ▶ Understand the HJB equation and the verification theorem
- ▶ See why CARA and CRRA utilities give tractable problems

Lecture 1: Dynamic Programming & HJB

- ▶ Controlled diffusions and admissible controls
- ▶ The dynamic programming principle
- ▶ The infinitesimal generator
- ▶ Deriving the HJB equation; boundary/terminal conditions

Lecture 2: Verification & Utilities

- ▶ The verification theorem: a smooth HJB solution IS the value function
- ▶ Viscosity solutions: when value functions are non-smooth
- ▶ CARA vs. CRRA vs. HARA
- ▶ Why CARA decouples the value function from wealth

- ▶ Solve the Merton problem under CARA, then CRRA
- ▶ Compare the two solutions side by side
- ▶ Recognize the recurring HJB $\rightarrow$ ansatz $\rightarrow$ ODE $\rightarrow$ solution pattern

Reading & Problem Set 4 (Python/PyTorch)

- ▶ Pham (2009) Ch. 3-4; Cartea-Jaikumar-Penalva Ch. 6
- ▶ PS4: Merton CRRA/CARA, Almgren-Chriss, HJB-residual check